

Placement Prep

Core CS Concepts Checklist · OS · CN · DBMS · OOP · System Design

1. OPERATING SYSTEMS

Processes & Threads

- Process vs Thread
- Process states (New, Ready, Running, Waiting, Terminated)
- PCB (Process Control Block)
- Context Switching
- User-level vs Kernel-level Threads
- Multithreading models (Many-to-One, One-to-One, Many-to-Many)

CPU Scheduling

- FCFS (First Come First Serve)
- SJF / SRTF (Shortest Job First / Remaining Time)
- Round Robin — Time Quantum impact
- Priority Scheduling (Preemptive & Non-preemptive)
- Multilevel Queue & Multilevel Feedback Queue
- Metrics — Throughput, Turnaround Time, Waiting Time, Response Time

Process Synchronisation

- Race Condition & Critical Section
- Peterson's Solution
- Semaphores (Binary & Counting)
- Mutex vs Semaphore
- Classic Problems
 - ■ Producer-Consumer
 - ■ Reader-Writer
 - ■ Dining Philosophers
- Monitors

Deadlocks

- Necessary Conditions (Coffman's — Mutual Exclusion, Hold & Wait, No Preemption, Circular Wait)
- Resource Allocation Graph (RAG)
- Prevention, Avoidance, Detection & Recovery
- Banker's Algorithm

Memory Management

- Contiguous Allocation — Fixed & Variable Partitioning
- Fragmentation (Internal & External), Compaction
- Paging — Page Table, TLB, Effective Access Time
- Segmentation
- Virtual Memory — Demand Paging
- Page Replacement Algorithms — FIFO, LRU, Optimal, LFU
- Thrashing & Working Set Model

File Systems

- File Allocation Methods — Contiguous, Linked, Indexed
- Directory Structure — Single-level, Two-level, Tree, Acyclic Graph
- Inode
- Free Space Management — Bit Vector, Linked List

Disk Scheduling

- FCFS, SSTF, SCAN, C-SCAN, LOOK, C-LOOK
- Disk Structure — Sector, Track, Cylinder

Inter-Process Communication (IPC)

- Pipes (Named & Unnamed)
- Message Queues
- Shared Memory
- Sockets
- Signals

System Calls & Kernel

- Types of System Calls — Process, File, Device, Info, Communication
- Monolithic vs Microkernel vs Hybrid
- User Mode vs Kernel Mode
- Interrupts & Traps

2. COMPUTER NETWORKS

OSI & TCP/IP Models

- 7 Layers of OSI — Physical, Data Link, Network, Transport, Session, Presentation, Application
- TCP/IP 4-layer model
- Functions of each layer
- PDU at each layer — Bit, Frame, Packet, Segment, Data

Physical Layer

- Transmission Media — Guided (Twisted Pair, Coaxial, Fiber) vs Unguided (Radio, Microwave, IR)
- Bandwidth, Throughput, Latency
- Encoding — NRZ, Manchester
- Multiplexing — TDM, FDM, WDM, CDM

Data Link Layer

- Framing, Error Detection & Correction
- ■ Parity, CRC, Checksum, Hamming Code
- Flow Control
- ■ Stop-and-Wait, Sliding Window (Go-Back-N, Selective Repeat)
- MAC — CSMA/CD (Ethernet), CSMA/CA (WiFi)
- ARP & RARP
- Switches vs Bridges vs Hubs

Network Layer

- IP Addressing — IPv4, IPv6
- Subnetting & CIDR
- NAT (Network Address Translation)
- Routing Algorithms
- ■ Distance Vector (RIP — Bellman-Ford)
- ■ Link State (OSPF — Dijkstra)
- ■ BGP (Path Vector)
- ICMP — Ping, Traceroute
- Fragmentation & Reassembly
- Routers vs Layer-3 Switches

Transport Layer

- TCP vs UDP — comparison table
- TCP
- ■ 3-Way Handshake (SYN, SYN-ACK, ACK)
- ■ 4-Way Termination (FIN, FIN-ACK, ACK)
- ■ Flow Control — Sliding Window
- ■ Congestion Control — Slow Start, AIMD, Fast Retransmit
- ■ Sequence & Acknowledgement Numbers
- ■ TCP Header fields
- UDP — connectionless, use cases
- Ports & Sockets
- Multiplexing & Demultiplexing

Application Layer Protocols

- HTTP / HTTPS — Methods (GET, POST, PUT, DELETE, PATCH), Status Codes, HTTP/1.1 vs HTTP/2 vs HTTP/3
- DNS — Resolution process, Record types (A, AAAA, CNAME, MX, NS), Recursive vs Iterative
- DHCP — DORA process
- FTP, SFTP, SMTP, IMAP, POP3
- WebSockets vs HTTP
- REST vs GraphQL vs gRPC (basics)

Network Security

- SSL / TLS — Handshake, Certificates, PKI
- Symmetric vs Asymmetric Encryption
- Firewalls — Packet Filter, Stateful, Application-layer
- Common Attacks — DDoS, Man-in-the-Middle, DNS Spoofing, ARP Poisoning
- VPN

Misc / Important

- CDN (Content Delivery Network)
- Load Balancing
- Proxy — Forward & Reverse
- Latency vs Bandwidth vs Throughput
- Socket Programming basics

3. DATABASE MANAGEMENT SYSTEMS (DBMS)

Fundamentals

- DBMS vs File System
- Architecture — 3-tier (External, Conceptual, Internal)
- Schema vs Instance
- Data Models — Relational, Hierarchical, Network, ER

ER Model

- Entity, Attribute (Simple, Composite, Derived, Multi-valued)
- Relationships — One-to-One, One-to-Many, Many-to-Many
- Weak Entity & Identifying Relationship
- ER to Relational Mapping

Relational Model

- Keys — Super, Candidate, Primary, Foreign, Composite, Alternate
- Relational Algebra — Select (σ), Project (π), Union, Intersection, Difference, Cartesian Product, Join
- Relational Calculus — Tuple & Domain
- Integrity Constraints — Domain, Entity, Referential

SQL

- DDL — CREATE, ALTER, DROP, TRUNCATE
- DML — INSERT, UPDATE, DELETE, SELECT
- DCL — GRANT, REVOKE
- TCL — COMMIT, ROLLBACK, SAVEPOINT
- Joins — INNER, LEFT, RIGHT, FULL OUTER, SELF, CROSS
- Aggregate Functions — COUNT, SUM, AVG, MIN, MAX
- GROUP BY, HAVING, ORDER BY
- Subqueries & Correlated Subqueries
- Views
- Indexes — Clustered vs Non-Clustered
- Stored Procedures & Triggers
- Window Functions — ROW_NUMBER, RANK, DENSE_RANK, LEAD, LAG

Normalization

- Functional Dependencies (FD)
- Closure of FD set, Minimal Cover
- 1NF, 2NF, 3NF, BCNF
- 4NF (Multi-valued dependency), 5NF
- Lossless Join & Dependency Preservation
- Denormalization — when & why

Transactions & ACID Properties

- Atomicity — all or nothing
- Consistency — DB constraints maintained
- Isolation — concurrent transactions don't interfere
- Durability — committed data persists
- Transaction States — Active, Partially Committed, Committed, Failed, Aborted

Concurrency Control

- Problems — Dirty Read, Lost Update, Unrepeatable Read, Phantom Read
- Isolation Levels — Read Uncommitted, Read Committed, Repeatable Read, Serializable
- Lock-Based Protocols — Shared & Exclusive Locks
- Two-Phase Locking (2PL) — Strict, Rigorous
- Timestamp-Based Protocol
- Deadlock in DBMS — Detection & Prevention
- MVCC (Multi-Version Concurrency Control)

Recovery

- Log-Based Recovery — Undo, Redo, Undo-Redo
- Checkpointing
- Shadow Paging
- ARIES Recovery Algorithm (concept)

Indexing & File Organisation

- Primary, Secondary, Clustering Index
- Dense vs Sparse Index
- B-Tree & B+ Tree (most important)
- Hash Indexing
- Sequential, Heap, Hash File Organisation

NoSQL (basics for interviews)

- Types — Document (MongoDB), Key-Value (Redis), Column (Cassandra), Graph (Neo4j)
- CAP Theorem — Consistency, Availability, Partition Tolerance
- BASE vs ACID
- When to choose SQL vs NoSQL

4. OBJECT-ORIENTED PROGRAMMING (OOP)

Core Pillars

- Encapsulation — data hiding, getters/setters, access modifiers
- Abstraction — abstract classes vs interfaces
- Inheritance — Single, Multi-level, Multiple (via interface), Hierarchical, Hybrid
- Polymorphism
- ■ Compile-time (Method Overloading)
- ■ Runtime (Method Overriding, virtual functions)

Classes & Objects

- Class vs Object
- Constructor — Default, Parameterised, Copy
- Destructor
- Static members & methods
- this keyword
- Object Lifecycle

Inheritance Deep Dive

- super / base class call
- Method Overriding rules
- Diamond Problem & how it's solved
- final / sealed classes
- instanceof operator

Interfaces & Abstract Classes

- Abstract class — partial implementation, cannot instantiate
- Interface — full contract, multiple inheritance
- When to use which
- Default methods in interface (Java 8+)

Important OOP Concepts

- Coupling & Cohesion
- Composition vs Aggregation vs Association
- Dependency Injection
- Constructor chaining
- Upcasting & Downcasting
- Covariant return types

SOLID Principles

- S — Single Responsibility Principle
- O — Open/Closed Principle
- L — Liskov Substitution Principle
- I — Interface Segregation Principle
- D — Dependency Inversion Principle

Design Patterns (know the concept + use-case)

- Creational — Singleton, Factory, Abstract Factory, Builder, Prototype
- Structural — Adapter, Decorator, Facade, Proxy, Composite
- Behavioral — Observer, Strategy, Command, Iterator, Template Method

Exception Handling

- try / catch / finally / throw / throws
- Checked vs Unchecked Exceptions
- Custom Exceptions
- Exception hierarchy

Memory Management in OOP

- Stack vs Heap allocation
- Garbage Collection concepts
- Memory Leaks
- Smart Pointers (C++) / References (Java)

5. SYSTEM DESIGN

Fundamentals & Estimation

- Scalability — Vertical vs Horizontal Scaling
- Back-of-envelope estimation — QPS, Storage, Bandwidth
- Latency Numbers Every Engineer Should Know
- Availability, Reliability, Maintainability
- SLA / SLO / SLI

High-Level Design (HLD) Components

- Client-Server Architecture
- Monolith vs Microservices — trade-offs
- API Gateway
- Load Balancer — L4 vs L7, Algorithms (Round Robin, Least Connections, IP Hash, Weighted)
- Reverse Proxy
- Message Queues & Event-Driven Architecture — Kafka, RabbitMQ
- CDN — Push vs Pull
- DNS & How a request flows end-to-end

Databases in System Design

- SQL vs NoSQL — when to pick
- Database Replication — Master-Slave, Master-Master
- Database Sharding — Horizontal Partitioning, Shard Key
- Consistent Hashing
- Database Indexing strategy
- Read Replicas
- CQRS (Command Query Responsibility Segregation)

Caching

- Cache vs DB — latency difference
- Cache Strategies
 - ■ Cache-Aside (Lazy Loading)
 - ■ Write-Through
 - ■ Write-Behind (Write-Back)
 - ■ Read-Through
- Eviction Policies — LRU, LFU, FIFO
- Distributed Cache — Redis, Memcached
- Cache Invalidation — the hard problem
- Cache Stampede / Thundering Herd

CAP Theorem & Distributed Systems

- CAP — Consistency, Availability, Partition Tolerance (pick 2)
- PACELC extension
- Strong vs Eventual Consistency
- Distributed Transactions — Two-Phase Commit (2PC), Saga Pattern
- Leader Election — Raft, Paxos (conceptual)
- Gossip Protocol
- Quorum

Reliability & Fault Tolerance

- Replication — Active-Passive, Active-Active
- Failover strategies
- Circuit Breaker Pattern
- Retry with Exponential Backoff & Jitter
- Bulkhead Pattern
- Health Checks & Heartbeats
- Chaos Engineering concept

API Design

- REST principles — Stateless, Uniform Interface, Idempotency
- HTTP Methods & Idempotency table
- Pagination — Offset vs Cursor
- Rate Limiting — Token Bucket, Leaky Bucket, Fixed Window, Sliding Window
- API Versioning
- GraphQL vs REST vs gRPC
- Webhooks vs Polling vs Long Polling vs SSE vs WebSockets

Blob/Object Storage & File Systems

- Object Storage — S3, GCS
- Block vs Object vs File Storage
- Distributed File Systems — HDFS concepts
- Chunking large files
- Deduplication & Compression

Search & Indexing

- Full-Text Search — Elasticsearch / Solr
- Inverted Index
- Fuzzy Search, Relevance Ranking

Common Design Problems (practice these)

- URL Shortener
- WhatsApp / Chat System
- Instagram / Twitter Feed (News Feed)
- YouTube / Netflix (Video Streaming)
- Uber / Ola (Ride Sharing)
- Google Drive / Dropbox (File Storage)
- Notification System
- Rate Limiter
- Search Autocomplete / Typeahead
- Distributed Cache (design Redis)
- Web Crawler
- Payment System

Observability & Monitoring

- Logging — structured logs, log aggregation (ELK)
- Metrics — Prometheus + Grafana
- Distributed Tracing — Jaeger, Zipkin
- Alerting & On-call